Project SHARP- Power Generation and Delivery System

Technical Report

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Change record

Issue	Date	Total pages	Page(s) affected	Description of change
0.0	24/02/2026	11	0–11	Adapt Initial Template
0.1	16/03/2026	27	3–15, 18–21, 24–27	Added preliminary chapters 1, 2 & 5
0.2	20/04/2026	30	13–16, 21–28	Continued Chapter 5, Finished requirements
0.3	29/04/2026	40	18–21, 24, 32	Added Simulation Ch. 2.3
0.4	12/05/2026	46	19–38	Reorganised Chapters, Added Ch. 4 & 6

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Chapter 1

Introduction

TBD

1.1 Background

Stellar Space Industries

Stellar Space Industries (SSI) is a Dutch aerospace company located in Noordwijk near European Space Agency (ESA)'s ESTEC facility. SSI specialises in high precision manufacturing capabilities and covers the entire product development cycle including design, machining, assembly, integration and testing. Their production facility is equipped with high-precision machining and testing equipment.

In collaboration with ESA, SSI is developing a novel electrodeless electric propulsion system for small satellites. This propulsion system aims to increase efficiency, lifespan and reliability compared to current state-of-the-art electric propulsion systems. SSI is currently in their second phase of development where they aim to design build and test the entire system in-house.

Presently, several subsystems of the propulsion system are being developed and tested concurrently. During the development of these subsystems, SSI is also looking for opportunities to apply the technology of each component in other applications. One of the subsystems that is currently ready for testing and application is SSI's Flow Control Unit (FCU) which is a high-precision valve that can control the mass flow of gaseous propellants.

Project SHARP

In order to find a relevant application for their FCU, SSI has initiated the Solid Hydrogen Aircraft Regulated Propulsion (SHARP) project. SHARP is a joint investigation between SSI and Solid Flow, a Dutch company specialising in the development of solid hydrogen storage systems. The goal of SHARP is to investigate the feasibility of hydrogen-fuelled drones that utilise solid-state hydrogen storage. In this collaboration, Solid Flow provides the cool gas generator (CGG), a system capable of producing gaseous hydrogen on demand from solid hydrogen storage. SSI contributes its expertise in flow control systems for electric propulsion, which may be adapted for hydrogen-based applications.

SHARP is conducted under the Luchtvaart in Transitie (LIT) program of the RVO of the Dutch government, which aims to advance sustainable aviation technologies. SHARP aims to contribute to the aviation energy transition by providing a scalable, weight and volume efficient energy source that is CO_2 and NO_x neutral. SHARP first aims to demonstrate the feasibility of the technology in drone applications, allowing future projects to scale the technology to larger aircraft and replace traditional jet fuel-based propulsion systems.

SHARP — Power Generation and Delivery System

This document considers the project plan for a subsystem of the SHARP project, specifically the development of the power generation and delivery system. This part of the SHARP project is conducted as a graduation internship. Throughout this document the part of SHARP that falls under the graduation internship will be referenced as SHARP — Power generation and feed System (SHARP-PS).

1.2 Project Results

For Stellar Space Industries, the primary goal for SHARP is to test and apply their FCU in a real world scenario and thus advance the FCU's technology readiness level (TRL) from 4 to 6. In addition to this, they aim to find a secondary market for their FCU in hydrogen based aviation compared to electric propulsion for satellites. Solid Flow has the same goals for their hydrogen CGG. The goal of SHARP-PS is to integrate the FCU and CGG and demonstrate the technologies and their compatibility in a relevant environment.

The relevant environment for the SHARP-PS project will be a hydrogen fuel cell that delivers power to an unmanned areal vehicle (UAV). The results of the SHARP-PS project will be the following:

- The design for a power generation and feed system that is integrated into a UAV.
- A test bench setup that validates the performance of this power generation and feed system.
- A research report that documents the design process and test results and assesses the feasibility of using this power system in UAV's.

If allowed by time, a UAV with the integrated power generation and feed system may be an additional result of the SHARP-PS project.

Requirements for the UAV are not described in this document. These are dependent on the mission scenario for the UAV which, as of the writing of this document, has not yet been determined. This mission scenario will be determined by Stellar Space Industries as soon as possible, after which the requirements for the UAV will be constructed. Requirements for the other results of the SHARP-PS project are as follows:

- Stellar Space Industries' FCU shall be included in the system.
- Solid Flow's hydrogen CGG shall be included in the system.
- A fuel cell shall be used to generate electricity from hydrogen.
- The system shall function without external power supply.
- The power density (power output over system weight) of the system must be the same or higher than that of state-off-the-art UAV power systems.
- Test results will be recorded and presented in a research report that is in accordance with 'Report writing for readers with little time' (?, ?)
- The research report shall answer the following research question: 'To what extent is a hydrogen fuel cell based power generation and feed system that integrates a Solid Flow's cool gas generator and Stellar Space Industries' flow control unit feasible for extending the mission time for UAV's'

The aforementioned research question will be subdivided into the following sub research questions:

RQ-1 What functional, performance, and safety requirements must the power generation and feed system meet to be integrated into a UAV?

RQ-2 What are the relevant performance characteristics of Solid Flow's cool gas generator and Stellar Space Industries' flow control system?

RQ-3 What flight-time improvement is theoretically achievable with hydrogen storage and fuel cells compared to state-of the-art UAV battery systems?

RQ-4 Which system architecture best balances safety, efficiency, weight, system complexity, and manufacturability for the power generation and feed system?

RQ-5 Which components (fuel cell, intermediate energy storage, power electronics) are most suitable based on the system requirements and design constraints?

RQ-6 What mechanical modifications are required to integrate the hydrogen storage, generator, fuel cell, and power electronics into a UAV airframe?

RQ-7 What electrical architecture ensures stable power delivery from the fuel cell system to state-of-the-art UAV propulsion systems and onboard electronics?

RQ-8 How does the integrated hydrogen-based power system perform during ground testing in terms of power output, stability, efficiency, and safety?

RQ-9 What technical risks, failure modes, or limitations affect the overall feasibility of implementing the system in operational UAV flights?

These research questions have been arranged into a research matrix where they have been assigned a research method and is specified what instruments or tools are used and what the output is of each research question. This research matrix is shown in Appendix A.

Chapter 2

Literature Review

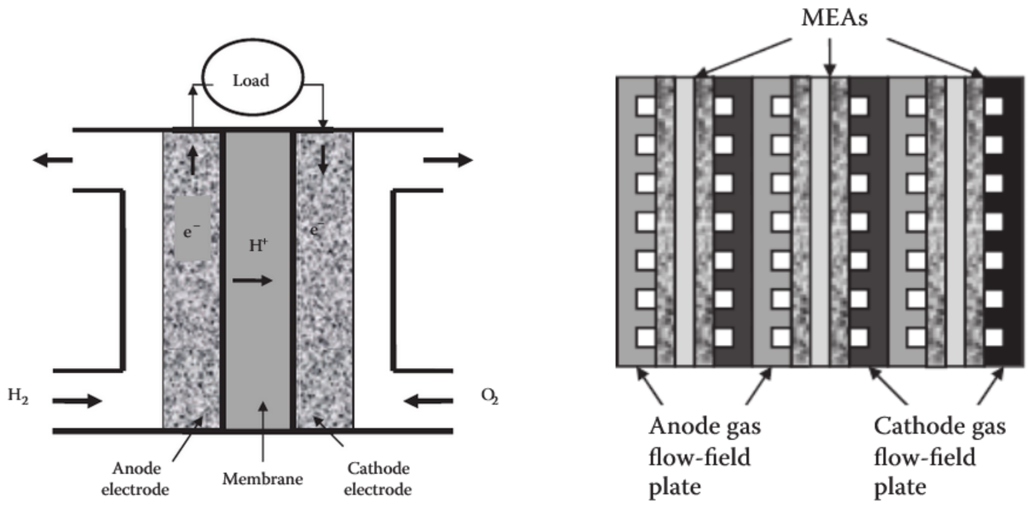
This chapter contains literature research that applies to the SHARP-PS project. The workings of fuel cells are explained in detail. A comparison is made that evaluates the effectiveness of energy storage as hydrogen tanks versus batteries. Several papers are described of other projects that are similar to the SHARP project. The results and design choices made during these projects might be of interest during the SHARP-PS project.

2.1 PEM Fuel Cells

Fuel cells are a technology that produce electricity from gases directly without combustion. There are many different types of fuel cells that run on different types of gases. These include solid oxide fuel cells (SOFC), direct methanol fuel cells (DMFC) and proton exchange membrane fuel cells (PEMFC). The PEMFC is used widely as it has a relatively high power density and a low operating temperature (Kim, 2014). These properties are especially beneficial for UAV power systems.

The book *Fuel Cells* by Revankar and Majumdar (2016) describes the working of the PEMFC in great detail. The PEMFC works by letting hydrogen undergo a series of electrochemical reactions. At the anode the hydrogen is split into protons (H^+) and electrons (e^-). The protons pass through the Proton Exchange Membrane (PEM), but the electrons can't pass through this membrane and must follow the external electrical circuit. The protons and electrons both arrive at the cathode where they react with oxygen to form water. This process is shown in a schematic diagram in figure 2.1a.

As a single fuel cell typically produces 0.7 volts, multiple cells are often connected in a stack. The membrane electrode assembly (MEA) is sandwiched between two conducting plates that have integrated flow channels for the hydrogen, oxygen and water. A schematic overview of a stack including three cells is provided in figure 2.1b.



(a) Single Fuel Cell

(b) Fuel Cell Stack

Figure 2.1: Fuel Cell Schematic Diagrams

As visible in figure 2.1a, hydrogen not only flows into the anode side of the fuel cell, but also out of that side. Simply leaving this side open to the air not only wastes a lot of unused hydrogen but also poses a major safety risk (Li, Wu, Cheng, Li, & Zhou, 2024). If this side of the anode is simply blocked off, impurities in the hydrogen and liquid water generated by the reaction accumulate over time, blocking the hydrogen supply. Instead, a purge valve can be installed that periodically purges the anode side and removes gas impurities and excess water. While this design is simple it still leads to periodic voltage drops and reduced power output. Li et al. introduces several circulation mechanisms to prevent this and enhance both safety and fuel utilisation. These circulation mechanisms include recirculation pumps, ejector valves or a combination of both. However both of these systems only become efficient for fuel cells operating above 80 kW, which is orders of magnitude higher than required for the fuel cell in this project.

2.2 Comparison of Energy Storage Methods

The SHARP project aims to extend the flight time of drones and possibly small aircraft in the future. It aims to do this by providing an alternative energy source that is more weight efficient than current solutions. To keep track of the progress and improvements that are made by the SHARP project the energy density of current solutions are analysed.

Different aspects of energy density play a role in aviation systems. Most important is the gravimetric energy density, which refers to the amount of stored energy per unit mass. For hydrogen this is 142 MJ/kg, nearly three times that of kerosene at 50 MJ/kg. However, volumetric energy density also plays a role and it refers to the amount of stored energy per unit volume. For hydrogen at atmospheric pressure this is only 0.01 MJ/L while for kerosene this is much higher at 35 MJ/L. For this reason, hydrogen is often stored in pressurised tanks or as a cryogenic liquid. This significantly increases the volumetric energy density as can be seen in table 2.1.

Fuel	Gravimetric [MJ/kg]	Volumetric [MJ/L]
Gasoline	46	34
Kerosene	50	35
Hydrogen (1 bar)	142	0.01
Hydrogen (350 bar)	142	3
Hydrogen (700 bar)	142	5
Hydrogen (liquid)	142	8.5

Table 2.1: Energy density of fuels

The gravimetric energy density of hydrogen itself may be very high, however pressurised storage vessels are quite heavy and thus their weight cannot be neglected. Generally, the hydrogen stored in a pressurised cylinder only contributes 2 to 5% of the total weight of the cylinder. This weight percentage is dependant on the cylinder shape and size. Intelligent Energy, a fuel cell supplier, has provided an overview of currently available hydrogen storage cylinders of different sizes from different suppliers (Intelligent Energy, 2022). This is shown in figure 2.2.

Hydrogen cylinders can now be compared with LiPo batteries, which are a popular energy storage solution for UAVs. From figure 2.2 it is clear that hydrogen cylinders become more weight efficient at larger sizes. LiPo batteries also come in different shapes and sizes, however they do not share this increase in efficiency with hydrogen cylinders. A comparison has been made the energy density of different systems at different storage sizes, this is shown in figure 2.3. The comparison includes some of the smaller Hydrogen cylinders from AMS CC from the comparison by Intelligent Energy, as well as some even smaller hydrogen tanks by MyH2. Several LiPo batteries have been included as well as batteries of some popular drones by DJI. A fuel cell efficiency of 40% was assumed when determining the energy capacity of the hydrogen cylinders.

The goal of this comparison is to find a break-even point where hydrogen cylinders become more efficient than batteries. Unfortunately, figure 2.3 does not give an exact point, however the first hydrogen cylinder that is more efficient than batteries has an energy capacity of 780 kJ, above this point hydrogen cylinders are more efficient than batteries. Noteworthy is that this is more than double the capacity of the DJI Inspire, a large consumer drone, at 350 kJ. This means that for doubling the flight time of this drone, it would require less mass to just add an extra battery than to replace its battery with a hydrogen system.

Figure 2.3 compares the gravimetric energy density of energy storage solutions, but ignores the weight of the system required to actually retrieve this energy. Figure 2.3 was made specifically for the context of the SHARP project where hydrogen is generated at a using a CGG. In order to decouple the constant rate of energy

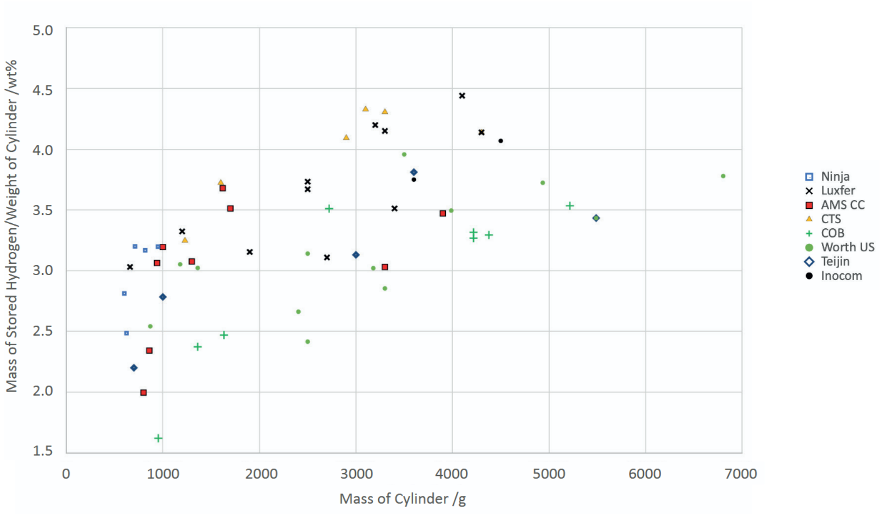


Figure 2.2: Gravimetric Energy Density comparison of hydrogen Cylinders by Intelligent Energy

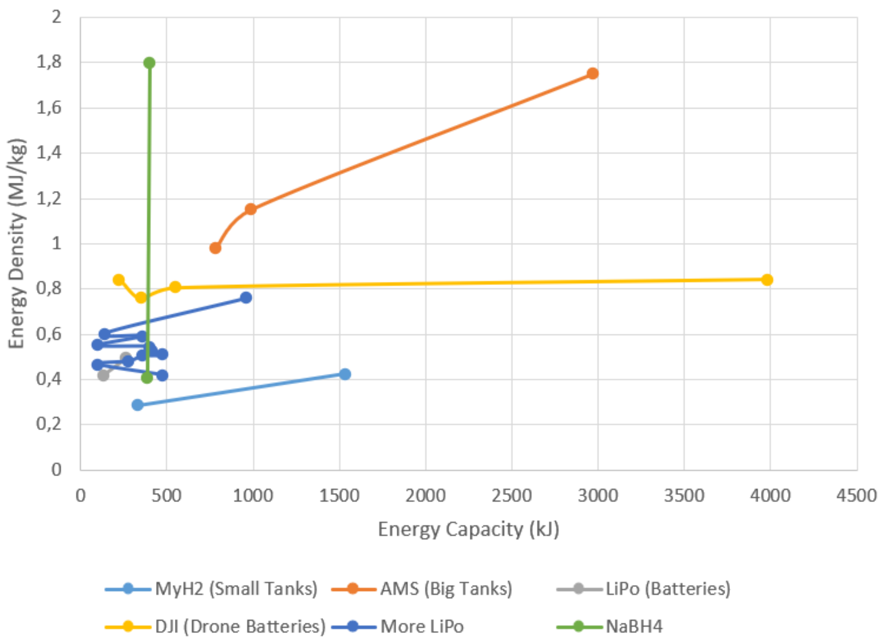


Figure 2.3: Gravimetric Energy Density comparison of different energy storage solutions

generation from the fluctuating energy consumption of the drone an energy buffer is required. Figure 2.3 allows comparison for using a battery or hydrogen cylinder as this buffer. In this specific case, the fuel cell system is required regardless of whether the energy buffer is implemented as a battery or as a hydrogen tank, and thus its weight does not play a role in their comparison.

However, when comparing using hydrogen cylinders with a fuel cell to using batteries without a fuel cell, the weight of the fuel cell must also be considered.

2.3 Hydrogen Drones

Several studies have been performed on the feasibility of hydrogen drones using PEMFCs. One notable example is van Benthem, de Boer, van der Vorst, and van Doorn (2020), their HYDRA project produced an 8kg hexacopter running on an 800W fuel cell for the Royal Netherlands Aerospace Centre (NLR). Their HYDRA-1B power system achieved and energy density of about 0.36 kWh/kg which is higher than that of commercial LiPo batteries at about 0.2 kWh/kg and still has significant potential for improvement. Their paper on the HYDRA project goes into detail about the drone design and part selection as well as hydrogen safety. This may provide a starting point for the design of the SHARP-PS project as the target scale of the drone and power system are similar.

Another example is De Wagter et al. (2021) who produced a Vertical Take-Off and Landing (VTOL) UAV called the Nederdrone, which runs on the same 800W fuel cell. In addition to stating the processes of fuel cell selection and drone design, they also go into detail about their hybrid power system using batteries to supply additional power during peak load.

Notable is that both systems use the same fuel cell by Intelligent Energy in combination with backup batteries. In addition to that, both projects are recent projects executed in the Netherlands, which may enable easy consultation if necessary. Stellar Space Industries has contacted NLR, who have expressed interest in the SHARP project.

Chapter 3

Requirements

From the mission scenario presented by Wren Aerospace et al. (discussed in section **??***) a set of system requirements is derived. From this mission scenario the system functions are formulated, which are decomposed into subfunctions. The system requirements are then allocated to these system functions.*

3.1 User Level Requirements

The user requirements defined by Stellar Space Industries, Solid Flow and Wren Aerospace are summarised below:

- The system shall include SSIs FCU
- The system shall include SolidFlows CGG
- The system shall integrate with WrenAerospaces UAV
- The system shall produce electricity from hydrogen
- The system shall have a power density greater than or comparable to UAVs

3.2 System Level Requirements

These user requirements are translated to system requirements in a structured manner. Each requirement will adhere to the following structure:

Type-ID	Requirement Statement (Category)
Proposed Approach	
Verification Approach	
Comment	

- Type: Type of requirement e.g. Performance Requirement (PERF)
- ID: SI no.
- Requirement Statement — a brief and unambiguous representation of the requirement
- Requirement Approach — provides information on the scope and how the requirement could be achieved
- Verification Approach — presents plan on how the specified requirement will be verified
- Category: Mandatory (M), Desirable (D), Optional (O)

Performance Requirements

PERF-1	The system shall be able to deliver 200 W continuous power	(M)
Proposed Approach	Ensure fuel cell meets power specifications from manufactures datasheet	
Verification Approach	Use wattmeter to verify	
Comment		

PERF-2	The system shall be able to supply loads varying from 140–220 W	(M)
Proposed Approach	Use a backup battery to provide power during peak loads	
Verification Approach	Use motor with propeller as variable load and wattmeter to verify	
Comment	Variable load profile will be dependant on mission profile. Determine representative load profile with Wren Aerospace.	

PERF-3	The system response time to load changes shall be identified	(M)
Proposed Approach	Use LiPo battery buffering	
Verification Approach	Verify using oscilloscope and step load changes	
Comment	Response time should be compatible with flight controller.	

Operational Requirements

OP-1	The system should be able to deliver power for 13–16 hours	(D)
Proposed Approach	Choose a CGG size appropriate according to calculations	
Verification Approach	Use mass flow sensor to verify fuel consumption of fuel cell	
Comment		

OP-2	The mass of the power generation and feed system should not exceed 5 kg	(D)
Proposed Approach	Divide mass budget over subsystems and select parts that fit said mass budget	
Verification Approach	Use scale to verify	
Comment		

OP-3	The system shall include a control system to regulate hydrogen mass flow	(M)
Proposed Approach	Implement feedback loop using mass flow sensor	
Verification Approach	Verify by step response testing	
Comment		

OP-4	The system shall support controlled startup and shutdown procedures (M)
Proposed Approach	Select a fuel cell with controller and/or develop startup/shutdown sequence logic
Verification Approach	Verify through repeated cycling tests
Comment	Most FCs come with a controller that does this

OP-5	The system shall provide minimal telemetry data including fuel content to the flight controller (M)
Proposed Approach	Integrate sensors and communication interface
Verification Approach	Verify through data logging
Comment	Translate the fuel content to a 'battery percentage' which can be communicated to the flight controller

OP-6	The system may provide additional telemetry data to the flight controller (O)
Proposed Approach	Integrate sensors and communication interface
Verification Approach	Verify through data logging
Comment	

Interface Requirements

INT-1	The volume of the energy storage systems should not exceed 2.5 L (D)
Proposed Approach	Design and select parts that fit in the given volume
Verification Approach	Size measurement using calipers or other appropriate measurement devices
Comment	Verify with Wren Aerospace if this volume is also valid for the fuel cell system

INT-2	The diameter of the energy storage system should not exceed 130 mm (D)
Proposed Approach	Design and select parts that fit in the given diameter
Verification Approach	Size measurement using calipers or other appropriate measurement devices
Comment	Verify with Wren Aerospace if this diameter is also valid for the fuel cell system

INT-3	The fluid interface with the CGG shall normal operating pressure of 3 bar (M)
Proposed Approach	Select parts compatible with this pressure
Verification Approach	Leakage test with compressed air
Comment	Exact value is dependant on buffer volume, 0.1 L —> 3 bar

INT-4	The fluid interface with the CGG shall accept a mass flow rate of 0.18–0.27 g/min (M)
Proposed Approach	Control fuel cell to use this mass flow rate of hydrogen using feedback control
Verification Approach	Verify using mass flow sensor
Comment	

INT-5	The electrical interface with the flight controller shall have a voltage of **XX** V(M)
Proposed Approach	Tune DC/DC converter to convert fuel cell output voltage to battery and flight controller voltage
Verification Approach	Verify using mass voltmeter
Comment	Probably 6S LiPo voltage (18–25.2 V). Confirm with Wren

INT-6	The system shall include standard electrical connectors compatible with the drone platform (D)
Proposed Approach	Select connectors per platform specification
Verification Approach	Visual and mechanical inspection
Comment	

Environmental Requirements

ENV-1	The system should be able to operate at speeds up to 21 m/s (D)
Proposed Approach	Verify fuel cell performance at appropriate atmospheric conditions
Verification Approach	
Comment	Translate to spec demands for fuel cell

ENV-2	The system shall operate at a temperature between -35 °C and 25 °C (D)
Proposed Approach	Select components rated for temperature range
Verification Approach	Verify in environmental chamber
Comment	Average temperature at 3 km (-4.5 °C) ±30 °C

ENV-3	The system shall operate at ambient pressures of 0.6–0.8 bar (D)
Proposed Approach	Validate fuel cell air supply and performance
Verification Approach	Test under reduced pressure conditions
Comment	Average pressure at 3 km (0.7 Bar) ±0.1 Bar

ENV-4	The system should tolerate relative humidity levels between **50–80%** (D)
Proposed Approach	Validate fuel cell datasheet and performance
Verification Approach	Verify under controlled humidity conditions
Comment	Typical Dutch weather, may be adjusted for all weather applications

Safety Requirements

SAFE-1	The system shall have a total hydrogen leakage lower than **XX** (M)
Proposed Approach	Design using COTS components rated for appropriate pressure
Verification Approach	Verify with leakage test
Comment	

SAFE-2	The sytem shall have a maximum burst pressure of **12?** bar (M)
Proposed Approach	Select parts compatible with this burst pressure
Verification Approach	Verify with manufacturer datasheets, potentially also a burst test with compressed air
Comment	Value should be normal operation pressure * safety factor (3 * 4 = 12)

SAFE-3	The system shall include overpressure protection (M)
Proposed Approach	Integrate pressure relief valve
Verification Approach	Verify using controlled pressure increase test
Comment	Value should be lower than burst pressure

SAFE-4	Expected hydrogen leakage should be vented along a safe path (M)
Proposed Approach	Connect pressure relief and purge output to tube that leads away from people
Verification Approach	Verify using visual inspection
Comment	

SAFE-5	The system should always be operated in appropriately ventilated areas or outside (M)
Proposed Approach	Always work outside when working with hydrogen
Verification Approach	Verify using visual inspection
Comment	

Chapter 4

Concept Design

This chapter presents the conceptual design process of the hydrogen-based power generation and feed system. Different system architectures are generated using a morphological analysis method and subsequently evaluated using a trade-off study. Based on this evaluation, the most suitable architecture is selected for further development.

4.1 System Objectives and Functional Requirements

The purpose of the power generation and feed system is to provide reliable electrical power using hydrogen generated from Solid Flow's cool gas generator. The generated hydrogen is supplied to a fuel cell, which converts the chemical energy into electrical energy.

The conceptual design of the system is driven by the following primary objectives:

- Provide stable electrical power to the load;
- Ensure safe handling and storage of hydrogen;
- Minimize system complexity and mass;
- Allow temporary energy buffering for peak load conditions;
- Enable controllable fuel cell operation;
- Maximize overall system efficiency.

To achieve these objectives, the system is divided into several key functional blocks:

- Hydrogen generation subsystem;
- Hydrogen storage and buffering subsystem;
- Fuel cell power generation subsystem;
- Electrical energy storage subsystem;
- Power and throttle control subsystem.

4.2 Morphological Analysis

A morphological analysis was conducted to systematically generate different system concepts. For each key system function, several implementation options were identified. By combining these options, multiple feasible system architectures can be created.

The resulting morphological chart is shown in Table 4.1.

System Function	Option 1	Option 2	Option 3
Fuel Cell Throttle	Hydrogen Flow Controller	Current Limiter	Relay
Energy Storage	Battery	Hydrogen Tank	–
Throttle Control	On-Off	PID	Fuzzy Logic

Table 4.1: Morphological chart of the proposed power generation and feed system

The morphological chart illustrates the design freedom available for the implementation of the system. Not all combinations are equally feasible; therefore, three representative architectures were selected for further evaluation.

4.3 Description of Proposed Architectures

Based on the combinations generated from the morphological chart, three candidate system architectures were developed. These architectures are summarized in Table 4.2.

	Architecture 1	Architecture 2	Architecture 3
Fuel Cell Throttle	Hydrogen Flow Controller	Relay	Current Limiter
Energy Storage	Hydrogen Tank	Battery	Battery
Throttle Control	Fuzzy Logic	On-Off	PID
Advantages	Simple Electronics	Simple Control System	Stable Power Delivery
Disadvantages	Heavy at small scale	Power Spikes	Complex Electronics
	No Backup Battery		

Table 4.2: Proposed system architectures

4.3.1 Architecture 1: Hydrogen Buffer-Based System

Architecture 1 uses a hydrogen tank as intermediate energy storage. Excess hydrogen produced by the sodium borohydride reactor is temporarily stored before being supplied to the fuel cell.

The fuel cell output is regulated using a hydrogen flow controller combined with a fuzzy logic control algorithm. This approach enables adaptive control of the hydrogen supply based on the load demand.

The main advantage of this architecture is the relatively simple electrical subsystem, since transient power demands can partially be compensated through hydrogen buffering. However, hydrogen storage tanks become relatively heavy and inefficient at small scales. Furthermore, the absence of an electrical backup battery reduces the system robustness during startup and transient conditions.

4.3.2 Architecture 2: On-Off Controlled Battery System

Architecture 2 uses a battery as the primary energy buffer. The fuel cell is controlled using a simple relay-based on-off strategy.

In this architecture, the fuel cell operates either at nominal output power or is completely switched off. The battery compensates for load transients and temporary mismatches between power generation and demand.

The simplicity of the control system is a major advantage of this architecture. However, the discontinuous operation of the fuel cell may introduce voltage fluctuations and power spikes, potentially reducing overall system stability and fuel cell lifetime.

4.3.3 Architecture 3: PID Controlled Battery System

Architecture 3 combines battery-based energy storage with active current limiting and PID-based throttle control.

The PID controller continuously regulates the fuel cell operating point in response to load variations, resulting in smoother power delivery and improved dynamic response. The battery acts as a supplementary energy buffer during transient operating conditions.

Compared to the previous architectures, this concept offers superior electrical stability and controllability. However, the increased control performance comes at the cost of higher electronic and software complexity.

4.4 Architecture Trade-Off and Selection

To objectively compare the proposed architectures, a trade-off analysis was performed. The concepts were evaluated on several criteria derived from the system objectives.

The evaluation criteria are:

- System complexity;
- Power stability;
- Scalability;
- Efficiency;
- Safety;
- Mass and volume;
- Reliability.

Each criterion was assigned a weighting factor based on its relative importance to the intended application. The architectures were subsequently scored on a scale from 1 to 5, where 5 represents the best performance.

Criterion	**Weight**	**Arch. 1**	**Arch. 2**	**Arch. 3**
Complexity	5	4	5	2
Power Stability	3	3	2	5
Efficiency	4	3	3	4
Safety	5	3	4	4
Mass/Volume	4	2	4	3
Reliability	4	3	3	4
Weighted Score		63	74	65

Table 4.3: Trade-off analysis of the proposed architectures

Based on the trade-off analysis, Architecture 2 achieved the highest overall score and was therefore selected for further development.

The selected architecture will therefore serve as the baseline concept for the detailed system design presented in the following chapters.

Chapter 5

Detailed Design

TBD

5.1 Simulation

A python simulation has been created that will simulate the gas flow of the hydrogen power system, which provides insight to the behaviour of this complex system. This section describes the goal of the simulation, how it works and the results it produced.

5.1.1 Goal

The goal of the SHARP-PS project is to develop a power system for a UAV that produces electricity from hydrogen using a fuel cell. This hydrogen will not be supplied by a hydrogen tank, but by a Cool Gas Generator (CGG) which will be developed by Solid Flow.

This system is quite complex and has many open variables that all influence each other. For example, both the hydrogen generation and consumption rate affect the pressure of the hydrogen supply system. The hydrogen consumption rate is in turn affected by the frequency and length of purge events and the power demand on the fuel cell. The hydrogen production rate is dependent on the size of the CGG, but the size of the CGG also influences the volume available for the hydrogen thus also affecting the pressure in a different way.

In order to grasp this complex system of variables, a python simulation has been created that simulates the gas flows of the entire system. This allows the effect of changing each variable to be observed, which will be a useful tool during the development of the system.

5.1.2 Process

The core of the simulation are a set of volumes that contain either hydrogen or electricity, and several processes that move content from one volume to another at a rate determined by the system properties. Figure 5.1 provides an overview of these volumes and processes, and the parameters each depends on.

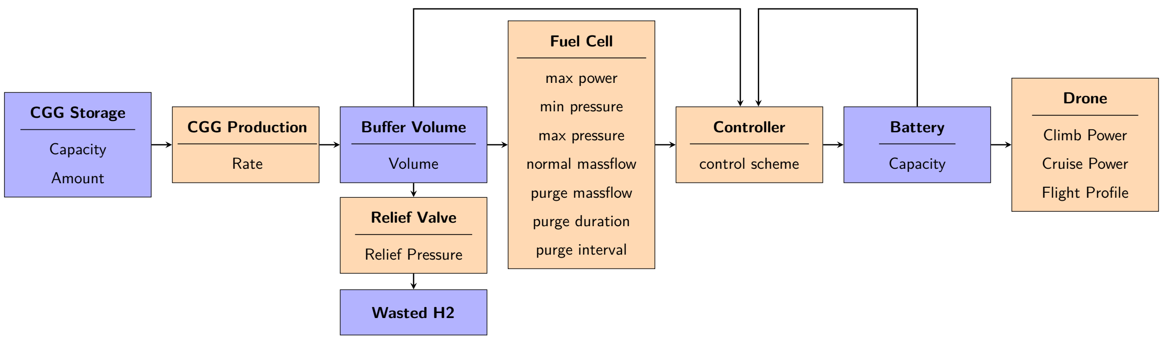


Figure 5.1: Simulation Diagram

5.1.3 Implementation

The initial parameters chosen for the simulation are based on the mission scenario by Wren Aerospace. These parameters are listed in table 5.1a. From these parameters, a preliminary fuel cell can be selected which will provide typical parameters for fuel cells in this range, visible in table 5.1b. The final selecting of the fuel cell to be used will be discussed later in chapter **??**. The simulation parameters may be adjusted at any time to suit the selected fuel cell.

The flight profile has not been determined yet, for the purposes of this simulation a climb of 1 hour has been chosen after which the drone will cruise for the rest of the mission.

Parameter	Value	Unit
CGG Capacity	160	g
Fuel Cell max Power	200	W
Climb Power	220	W
Cruise	160	W
Flight time	14	h

(a) Based on the Mission Scenario by Wren Aerospace

Parameter	Value	Unit
Min pressure	0.35	Bar
Max pressure	0.5	Bar
Normal massflow	0.263	g/min
Purge massflow	0.545	g/min
Purge duration	1–2	s
Purge interval	30–90	s

(b) Derived from the selected Fuel Cell

Parameter	Value	Unit
CGG Amount	1	—
CGG rate	0.21	g/min
Buffer Volume	0.1	L
Relief Pressure	4	Bar
Battery Capacity	500	kJ

(c) Chosen

Table 5.1: Simulation Parameters

The purpose of the controller is to limit the hydrogen consumption when the pressure in the buffer volume gets too low or when the battery is fully charged. For this purpose both an On—Off controller and a PID controller have been tested.

5.1.4 Results

Running the simulation with all parameters unchanged and as stated in the previous section produces the graph in figure 5.2. The pink line represents the power draw from the drone, this illustrates most clearly what phase of the mission is currently happening. Takeoff happens around half an hour into the simulation, then a 1 hour climb with high power demand after which the drone cruises for the rest of the mission, using less power. As a range was given for both climb and cruise power the simulation picks a random value in the range each timestep, hence this line appears very thick.

The red line represents the content of the battery. It starts empty and begins to charge, when it is at 50% the drone starts liftoff. This uses more power than the fuel cell can provide, therefore the battery supplies the additional power and thus discharges over time. When the drone switches to cruising the battery can charge again slowly. Note how the battery discharges rapidly around the 14 hour mark. This is because the CGG has run out (indicated by the yellow line) and the fuel cell can no longer provide any power. At this point, all the power is delivered by the battery until it is at 10%, then the mission is at an end or a new CGG is ignited in case multiple CGG's were brought on the mission.

The green, blue and cyan lines indicate pressure in the buffer volume, whether a purge event is happening and how much power the fuel cell produces respectively. At the scale of figure 5.2 their behaviour is poorly visible. Figure 6.2a provides a zoomed in view around the moment that a purge event is happening, there the behaviour of the three lines is better visible. Note however, that some lines have been excluded from this view and the scales have been altered for better visibility.

The blue line is at zero when no purge event is happening and jumps to one during a purge event. In the middle of the graph a purge event of around two seconds is visible. During the purge event the fuel cell uses hydrogen at an increased rate. Consequently, the pressure in the buffer volume drops to supply this extra hydrogen which can be seen in the green line. The controller notices this pressure drop and limits the power output of the fuel cell thereby limiting the hydrogen consumption, this can be seen by the cyan line dropping to zero.

After the purge event, the fuel cell is still limited, allowing the pressure in the buffer volume to build up again. When this pressure has been largely recovered, the fuel cell is allowed to produce power again. The reduction in power output from the fuel cell is compensated by the battery, which is visible in the red line.

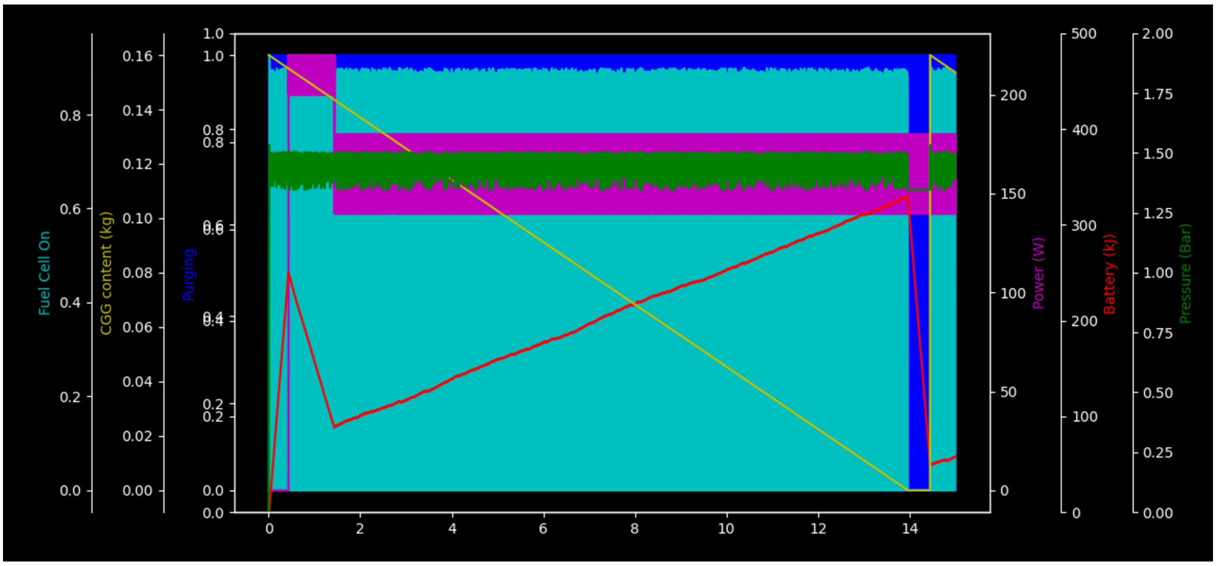


Figure 5.2: Simulation Result

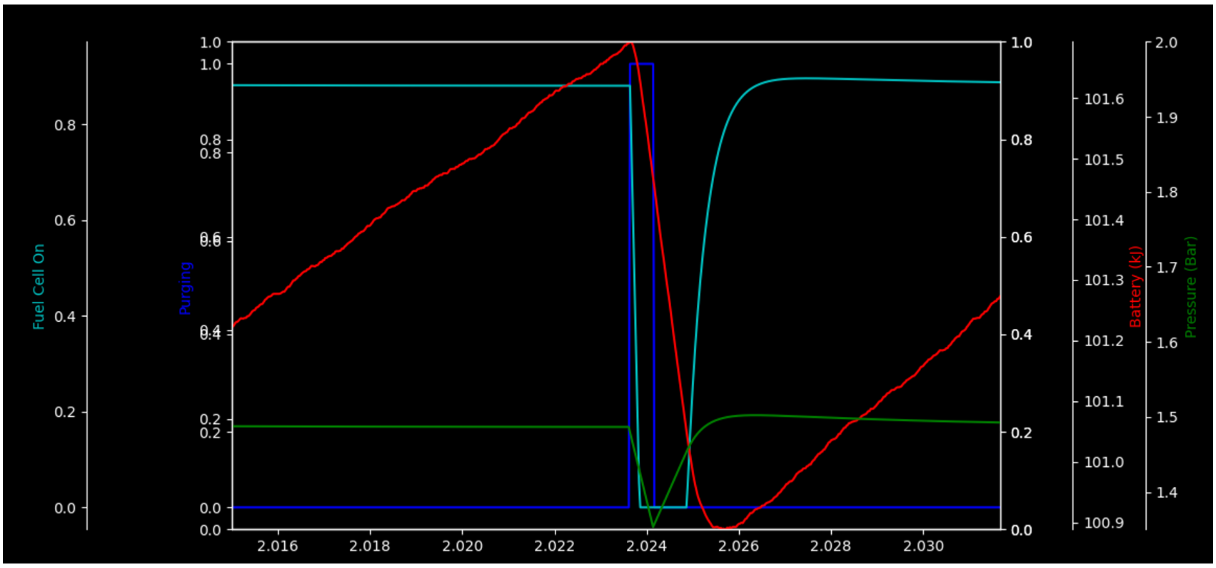


Figure 5.3: Purge Event

Many of the parameters used in this simulation were altered to investigate their effect on the overall system behaviour. These results were used during the design of the hydrogen power system, and as such will be discussed further in chapter 5.2.

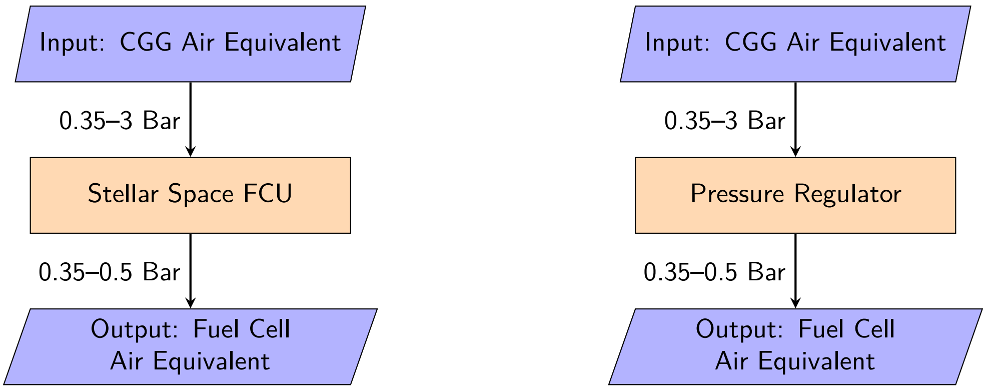
The simulation results will also need to be verified against real-world test results. For this and other purposes a compressed air system will be developed that will perform the gas flow of the hydrogen power system but with compressed air instead of hydrogen. This air-equivalent system will be discussed throughout the following chapters. Its results will be compared with the results of this simulation in chapter 6.

5.2 Air Equivalent Test Setup

TBD

5.2.1 Fuel Supply Regulator

TBD

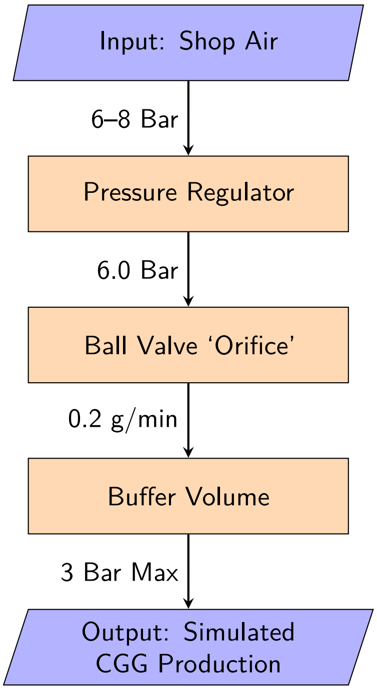


(a) FCU (b) Pressure Regulator

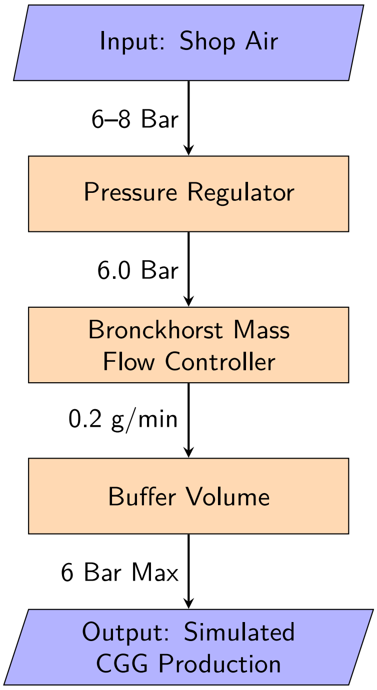
Figure 5.4: Compressed Air Equivalent

5.2.2 Cool Gas Generator

TBD



(a) Ball Valve



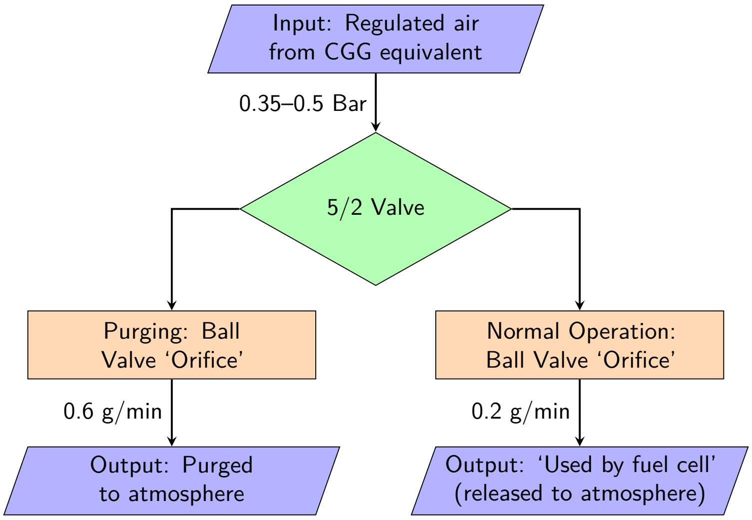
(b) Bronckhorst

Figure 5.5: CGG Compressed Air Equivalent

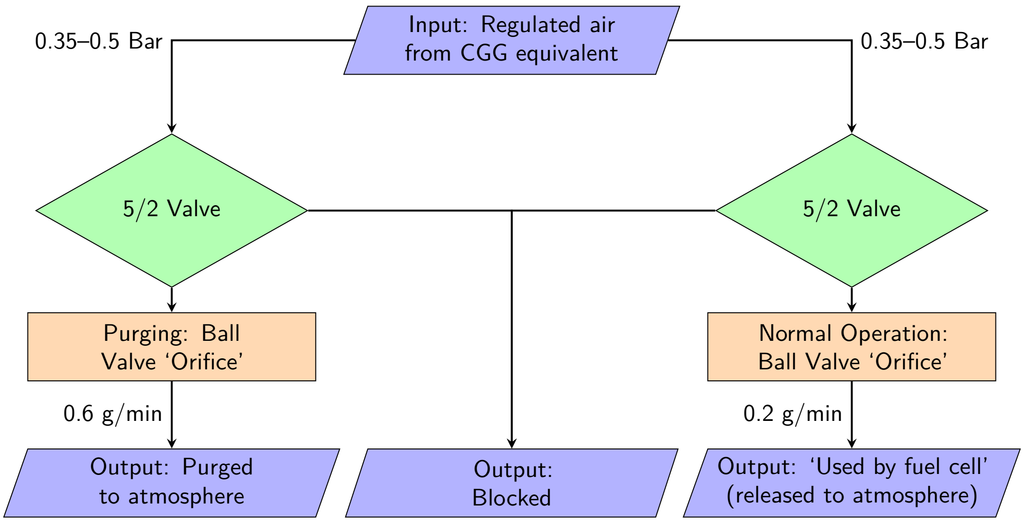
5.2.3 Fuel Cell

TBD

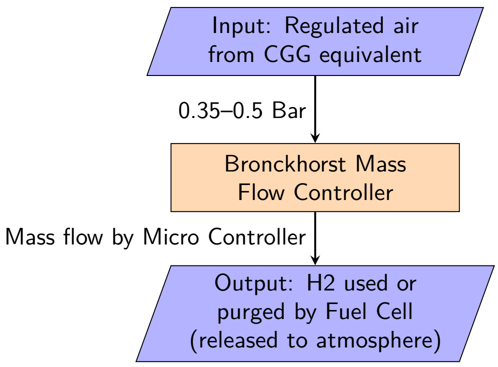
Test setup	Capabilities	Downsides
Ball Valve Single Switch	Test uncontrolled system behaviour	Mass flow is only accurate when input pressure is constant
Ball Valve Dual Switch	Test On–Off controlled behaviour	Mass flow is only accurate when input pressure is constant
Bronchorst Valve	Test PID controlled behaviour	Expensive



(a) Ball Valve, Single Switch



(b) Ball Valve, Dual Switch



(c) Bronckhorst

Figure 5.6: Fuel Cell Compressed Air Equivalent

5.3 Hydrogen Test Setup

Name	H-100	H-200	H2Gatech
Image			
Power	100 W	200 W	100 W
Price	€ 775	€ 1164	€ 715
Link	alibaba	alibaba	alibaba
Shipping	±1 month	±1 month	5–20 Days

Chapter 6

Test Results

TBD

Goals of various tests:

- Compressed Air Equivalent Test
  - Explore unknown risks
  - Verify expected system behaviour
  - Verify simulation results
  - Compare control strategies
- FCU Pressure test
  - Verify consistency of FCU output pressure
- FCU mass flow test
  - Verify consistency of FCU output mass flow
  - Verify leakage of FCU
- Gas supply test
  - Verify consistency of gas pressure during purge events
  - Verify consistency of gas mass flow during purge events
  - Verify total leakage of gas supply system
- Fuel Cell test
  - Verify the fuel cell is working normally
  - Verify the fuel cell output voltage
  - Verify the fuel cell output power
  - Verify the fuel cell fuel consumption
- Propulsion test
  - Verify the maximum thrust
  - Verify the maximum power draw
  - Verify the relation between input power and thrust output
- Hybrid Battery test
  - Verify the battery cannot back-feed power into the fuel cell
  - Verify output power of hybrid system
  - Verify output voltage of hybrid system
- Integration test
  - Verify all individually tested components behave as expected when integrated into one system
  - Verify relation between fuel consumption and thrust output

6.1 Compressed Air Equivalent System

The compressed air equivalent system performs the same gas flows as the hydrogen part of the power generation and feed system. This system will allow analysis of the flow dynamics of the system without the use of hydrogen, allowing for a cheaper and safer system that is faster to realise. In this chapter, an overview is given of the system and the tests performed with it. The results of these tests are reviewed and compared with those of the python simulation.

6.1.1 Testing Goals

The main goal of the compressed air equivalent tests is to explore the flow dynamics of the system and verify that they match expected behaviour. This includes comparing results with the python simulation and verifying that using the internal volume of the CGG as a buffer is sufficient to ensure stable flow to the fuel cell during purge events.

Additionally, the compressed air tests will be used to gain experience with the fluidics hardware and explore risks like leakage before introducing hydrogen.

6.1.2 System Overview

The hydrogen part of the power generation and feed system consists of three main components: the fuel cell, the flow control unit, and the cool gas generator. In the air equivalent system, the fuel cell and CGG are replaced by subsystems that let through air at the same rate and pressure as the fuel cell and CGG would. The flow control unit is used as-is, controlling the flow of air instead of hydrogen. The system is depicted in Figure 6.1.

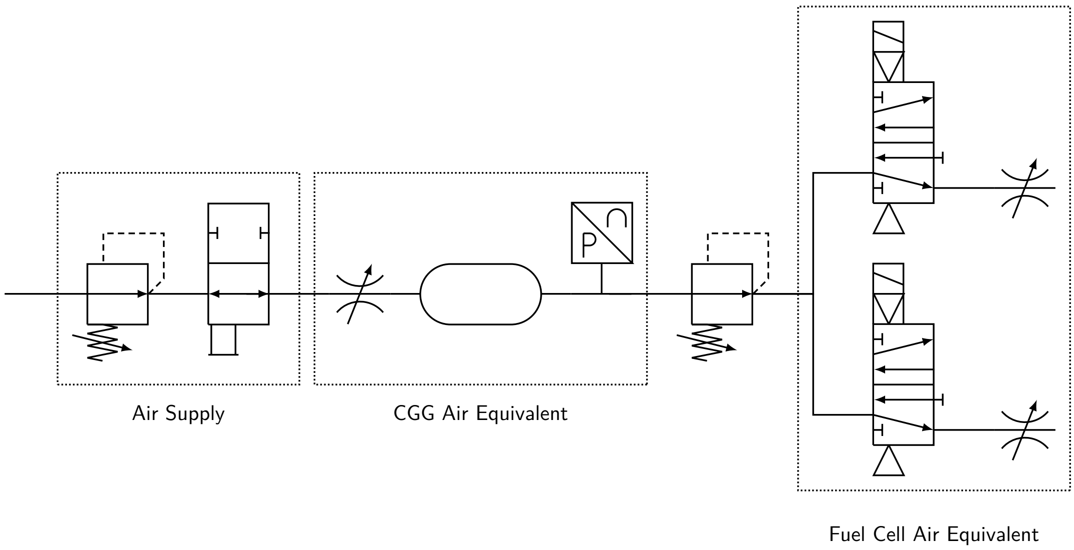


Figure 6.1: Fluidics of the Air Test Setup

CGG Equivalent

The CGG produces hydrogen at a fixed rate regardless of output pressure. To simulate this, an orifice has been sized to the expected mass flow of the CGG. To ensure that this flow rate does not change with pressure, sonic flow must occur in the orifice. This occurs when the upstream pressure is at least 2 times the downstream pressure. The maximum expected CGG output pressure is 4 Bar, so the upstream pressure is set to 8 Bar to ensure sonic flow. The hydrogen that the CGG produces is stored in the internal volume of the CGG, which is approximately 0.1 L. Therefore, the air system includes a buffer volume of 0.1 L to simulate the CGG storage. The CGG equivalent also includes a pressure sensor to measure the pressure inside the buffer volume.

Fuel Cell Equivalent

The fuel cell normally uses hydrogen at a rate that depends on power draw and at an increased rate during purge events. For the air tests, the 'power draw' is assumed to be a fixed value of 200 W, which corresponds to a hydrogen mass flow of approximately 0.18 g/min. During purge events, the hydrogen mass flow is expected to increase to approximately 0.27 g/min. To simulate this with compressed air, two orifices have been sized to these flow rates at a pressure of 1 Bar, which corresponds to the operating pressure of the fuel cell. Solenoid valves are used to switch between the two orifices to simulate purge events.

As the fuel cell requires an input pressure of exactly 1 Bar, but the CGG volume can fluctuate between 1 and 4 Bar, a pressure regulator is used to ensure the fuel cell equivalent always receives air at 1 Bar. In the hydrogen system, this function will be performed by Stellar Space Industries' flow control system. However, as this system is not yet available, a commercial off-the-shelf pressure regulator is used for the air tests.

Control System

TBD

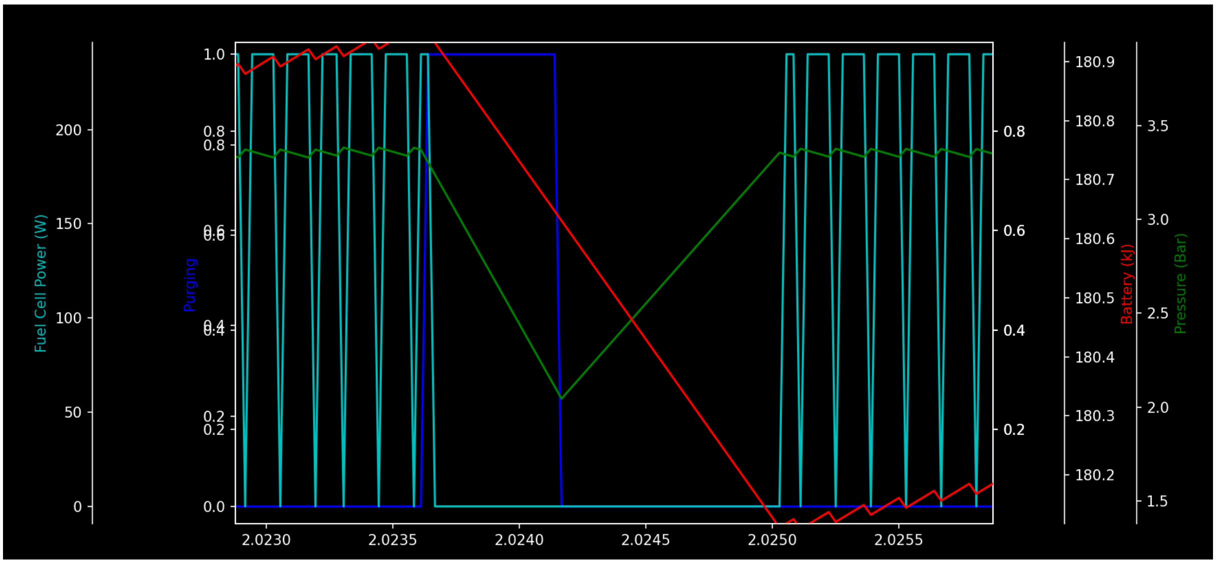
6.1.3 Results

Figure 6.2b show the pressure in the CGG buffer volume and the flow rate from the CGG to the fuel cell. Figure 6.2a shows the results of the python simulation during a similar time frame. The simulation also shows the pressure in the CGG buffer volume in green, which can be directly compared to the measure pressure in the air test. The simulation does not directly show the same flow rate. However, it does show the state of the fuel cell from which a flow rate can be inferred. The blue line in the simulation shows when the fuel cell is in purge mode, which corresponds to the higher flow rate in the air test. The cyan line shows when the fuel cell is in normal mode, which corresponds to the lower flow rate in the air test.

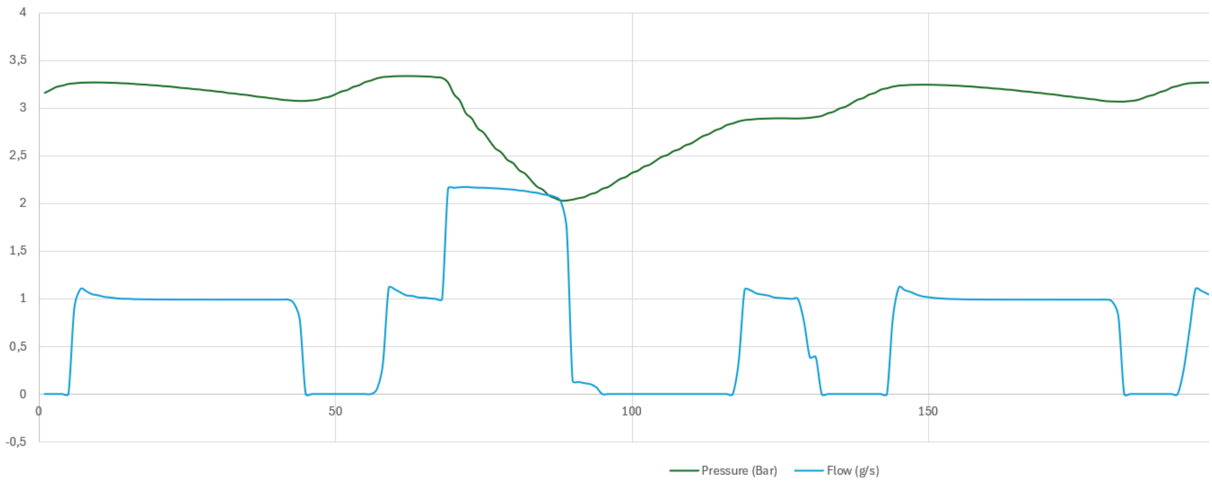
Note that the graph representing the flow rate of the air system has been scaled to improve visibility, this means that its units are no longer correct.

During normal operation, the fuel cell consumes slightly more than the CGG produces, which causes the pressure in the buffer to decrease. When the pressure drops too low, the fuel cell is temporarily disabled to allow the buffer to fill up again. This behaviour is seen in both the simulation and the test results. The real system reacts a bit slower to the pressure changes than the simulation, which leads to slightly larger pressure fluctuations. However, the overall behaviour is very similar, which gives confidence that the simulation is accurately capturing the flow dynamics of the system.

When a purge event occurs, the fuel cell equivalent briefly consumes a lot more air, which causes a sharp drop in pressure. This too occurs both in the simulation and in the test results. In both systems the pressure drops from around 3.4 Bar to around 2 Bar. After the purge event, both systems disable the fuel cell to fill up the buffer again.



(a) Simulated Purge Event



(b) Measured Air Purge Event

Figure 6.2: Comparison of Simulation results with Compressed Air system

6.2 Hydrogen Test

TBD

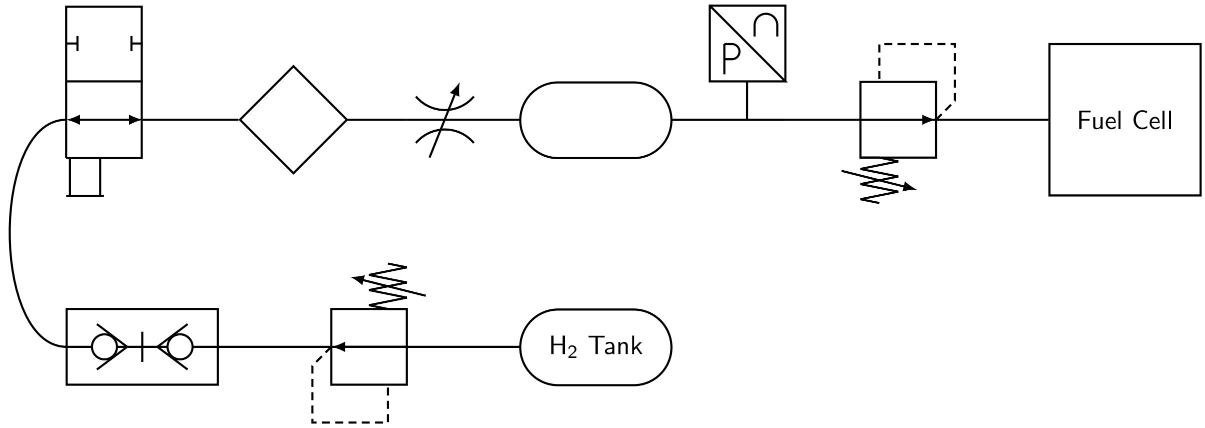


Figure 6.3: Hydrogen Test Setup

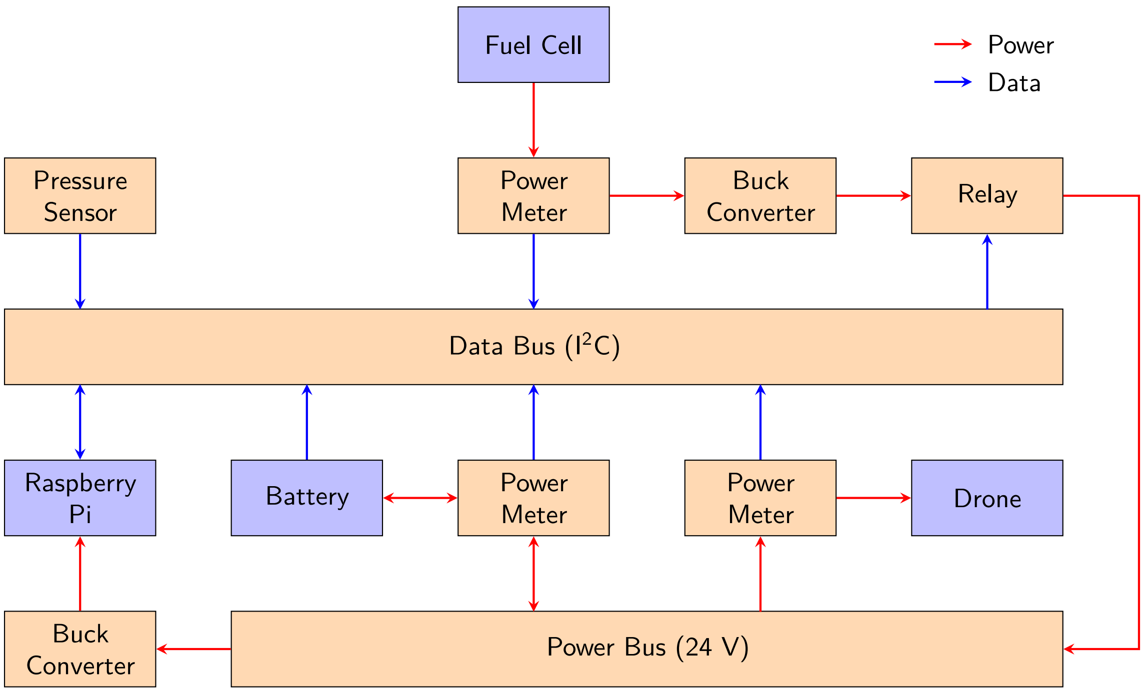


Figure 6.4: Electronics Test Setup

Component	Pin	Connector	Connects to	Connector	Wire
Pressure Sensor	Vcc	Attached wire	24V	Terminal Block	Loose Wire
	Data	Attached wire	4–20 mA Converter	Terminal Block	included
4–20 mA Converter	+	Terminal block	Pressure Sensor	Attached Wire	included
	-	Terminal block	GND	Terminal Block	Loose Wire
	Analog Volt	DuPont (f)	ADC	DuPont (m)	included
ADC	Analog Volt	DuPont (m)	4–20 mA Converter	DuPont (f)	included
	Analog Volt	DuPont (m)	Battery	JST-XH 7 (f)	JST-XH 7 (m) – DuPont (m) DuPont (ff)
	Analog Volt	DuPont (m)	—		
	Analog Volt	DuPont (m)	—		
	I^{2}C	DuPont (f)	I^{2}C Bus	DuPont (m)	included
I^{2}C Bus	I^{2}C	DuPont (m)	ADC	DuPont (f)	included
	I^{2}C	DuPont (m)	Power Meter	DuPont (m)	DuPont (ff)
	I^{2}C	DuPont (m)	Power Meter	DuPont (m)	DuPont (ff)
	I^{2}C	DuPont (m)	Power Meter	DuPont (m)	DuPont (ff)
	I^{2}C	DuPont (m)	—		
	I^{2}C	DuPont (m)	—		
	I^{2}C	DuPont (m)	—		
	I^{2}C	DuPont (m)	—		
	I^{2}C	DuPont (f)	Raspberry Pi	DuPont (m)	included
Fuel Cell	Power Out	XT60 (f)	Power Meter 1	Terminal Block	XT60 (m-)
Power Meter 1	Power In	Terminal Block	Fuel Cell	XT60 (f)	XT60 (m-)
	Power Out	Terminal Block	Buck Converter 1	Terminal Block	Loose Wire
	I^{2}C	DuPont (m)	I^{2}C Bus	DuPont (m)	DuPont (ff)
Buck Converter 1	Power In	Terminal Block	Power Meter 1	Terminal Block	Loose Wire
	Power Out	Terminal Block	Relay	Terminal Block	Loose Wire
Relay	Common	Terminal Block	Buck Converter 1	Terminal Block	Loose Wire
	NO	Terminal Block	24V	Terminal Block	Loose Wire
	Control	PCB Hole	Raspberry Pi	DuPont (m)	DuPont (mf)

Continued on next page...

Component	Pin	Connector	Connects to	Connector	Wire
Power Meter 2	Power In	Terminal Block	Battery	XT60 (f)	XT60 (m-)
	Power Out	Terminal Block	24V	Terminal Block	Loose Wire
	I^2C	DuPont (m)	I^2C Bus	DuPont (m)	DuPont (ff)
Battery	Power	XT60 (f)	Power Meter 2	Terminal Block	XT60 (m-)
	Balance Leads	JST-XH 7 (f)	ADC	DuPont (m)	JST-XH 7 (m) – DuPont (m) DuPont (ff)
Power Meter 3	Power In	Terminal Block	ESC	Attached Wire	included
	Power Out	Terminal Block	24V	Terminal Block	Loose Wire
ESC	Power In	Attached Wire	Power Meter 3	Terminal Block	included
	3 Phase Out	BLDC Plug (f)	BLDC	BLDC Plug (m)	included
	Throttle	DuPont (f)	Raspberry Pi	DuPont (m)	included
	BEC	DuPont (f)	—		
	Active Break	DuPont (f)	—		
BLDC	3 Phase In	BLDC Plug (m)	ESC	BLDC Plug (f)	included
Buck Converter 2	Power In	Terminal Block	24V	Terminal Block	Loose Wire
	Power Out	Terminal Block	Raspberry Pi	DuPont (m)	DuPont (f-)
Raspberry Pi	5V In	DuPont (m)	Buck Converter 2	Terminal Block	DuPont (f-)
	I^2C	DuPont (m)	I^2C Bus	DuPont (f)	included
24V Bus	Power	Terminal Block	Pressure Sensor	Terminal Block	Loose Wire
	Power	Terminal Block	Relay	Terminal Block	Loose Wire
	Power	Terminal Block	Power Meter 2	Terminal Block	Loose Wire
	Power	Terminal Block	Power Meter 3	Terminal Block	Loose Wire
	Power	Terminal Block	Buck Converter 2	Terminal Block	Loose Wire

Table 6.1: Electrical Connections

Chapter 7

Conclusion

TBD

Chapter 8

Recommendations

TBD

Appendix A

Research Matrix

RQ-1	Requirements
Description	What functional, performance, and safety requirements must the power generation and feed system meet to be integrated into a UAV?
Research Method	Literature review and stakeholder analysis
Instruments Tools	Scientific literature, hydrogen and UAV standards, requirement templates
Output Data	System requirements specification

RQ-2	Characteristics
Description	What are the relevant performance characteristics of Solid Flow's cool gas generator and Stellar Space Industries' flow control system?
Research Method	Technical analysis and literature study
Instruments Tools	Supplier datasheets, technical documentation, prior test data
Output Data	Performance characteristics overview

RQ-3	Flight-Time Improvement
Description	What flight-time improvement is theoretically achievable with hydrogen storage and fuel cells compared to state-of the-art UAV battery systems?
Research Method	Analytical modelling and comparison
Instruments Tools	Energy density calculations, MATLAB/Excel models
Output Data	Theoretical flight-time estimation

RQ-4	**System Architecture**
Description	Which system architecture best balances safety, efficiency, weight, system complexity, and manufacturability for the power generation and feed system?
Research Method	Conceptual design and trade-off analysis
Instruments Tools	Morphological chart, decision matrix
Output Data	Selected system architecture

RQ-5	**Component Selection**
Description	Which components (fuel cell, intermediate energy storage, power electronics) are most suitable based on the system requirements and design constraints?
Research Method	Component selection study
Instruments Tools	Supplier datasheets, selection criteria tables
Output Data	Selected component list with justification

RQ-6	**Mechanical Integration**
Description	What mechanical modifications are required to integrate the hydrogen storage, generator, fuel cell, and power electronics into a UAV airframe?
Research Method	Mechanical design
Instruments Tools	CAD software, mass and tolerance analysis
Output Data	Mechanical design drawings and mass breakdown

RQ-7	**Electrical Architecture**
Description	What electrical architecture ensures stable power delivery from the fuel cell system to state-of-the-art UAV propulsion systems and onboard electronics?
Research Method	Electrical system design
Instruments Tools	Electrical schematics, simulation tools
Output Data	Electrical architecture and schematics

RQ-8	**System Performance**
Description	How does the integrated hydrogen-based power system perform during ground testing in terms of power output, stability, efficiency, and safety?
Research Method	Experimental testing
Instruments Tools	Hydrogen supply, sensors, data acquisition system
Output Data	Measured performance and test results

RQ-9	**Feasibility and Risk**
Description	What technical risks, failure modes, or limitations affect the overall feasibility of implementing the system in operational UAV flights?
Research Method	Risk analysis and evaluation
Instruments Tools	Failure mode effect analysis (FMEA), test observations, design review
Output Data	Feasibility assessment and risk overview

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